Content-Guided Convolutional Neural Network for Hyperspectral Image Classification

Qichao Liu, *Student Member, IEEE*, Liang Xiao, *Member, IEEE*, Jingxiang Yang, *Member, IEEE*, and Jonathan Cheung-Wai Chan

***Abstract*— Convolutional neural networks (CNNs) are of great interest and have demonstrated remarkable performance in hyperspectral images (HSIs) classification. However, due to the current configuration of the convolution layers with a fixed kernel shape, regular CNNs are inherently limited in modeling the diverse land-cover structures, particularly in the cross-classes edge regions, where irregular class boundaries would lead to high classification errors. To address this issue, we propose a content-guided CNN (CGCNN) for HSI classification. Compared with the shape-fixed kernel in the traditional CNN, the proposed content-guided convolution adaptively adjusts its kernel shape according to the spatial distribution of land covers. The content pattern is reflected by a latent guide map automatically learned from HSI. Such content-adaptive kernel with CGCNN could suppress the irregularity and unexpected features in class boundaries and, thus, improve the feature learning in cross-classes regions. Based on the content-guided convolution, a novel guided feature extraction unit (GFEU) is constructed for spectral–spatial feature learning of HSI. Finally, the CGCNN classification framework is established by stacking multiple GFEUs with dense connection, which is helpful for mitigating the gradient vanishing and increasing the robustness to overfitting. Extensive experiments on several HSIs demonstrate that the proposed approach possesses great details' preserving ability and its performance outperforms other state-of-the-art methods.**

***Index Terms*— Content-guided convolution, convolutional neural network (CNN), guided kernel, hyperspectral image (HSI) classification, latent guide map (LGM).**

I. Introduction

HYPERSPECTRAL images (HSIs) are of 3-D cubes containing both rich spatial and spectral information, which helps to distinguish the materials on the pixel level. Therefore, HSI has been widely applied to many fields, including environmental monitoring, geological exploration, and target detection [1]–[4]. The most important technique for these applications is to classify every pixel in the HSI.

In the past decades, many supervised HSI classification methods have been proposed. From statistical models [5], [6] to feature representation-based methods [7]–[9], HSI classification has become a targeted research in the remote sensing field. Assuming that land covers are separable linearly in hyperspectral space, many typical linear classification methods have been applied to HSI, such as linear regression (LR) [10], support vector machine (SVM) [11], [12], linear discriminant analysis (LDA) [13], and principal component analysis (PCA) [14]. Due to the light-scattering mechanisms in surface reflectance, nonlinear classification methods, such as kernel SVM (KSVM) [15], kernel extreme learning machine (KELM) [16], and multiple kernel learning (MKL) [17], have shown more promising results. However, mixed pixels due to the limited resolution and the complex imaging process often lead to the phenomenon that different materials have similar spectra and the same materials have dissimilar spectra. Hence, spectral information alone is often not enough to classify the HSI accurately. To improve the classification performance, researchers have exploited the spatial information of HSI. For instance, the aggregation attribute of local homogeneous pixels has been utilized to boost HSI classification, such as the Markov random fields (MRFs) [18]–[20], object-based voting [21], and relearning [20], [21]. The local spatial patterns of HSI have been also investigated, such as the Gabor features [18], morphological features [22], and textural features [23]. At present, the spectral–spatial joint classification methods have become the mainstream in HSI classification.

However, when dealing with different types of HSIs, most traditional methods have inherent limitations because of the need for manual tuning of empirical parameters. Due to the ability in extracting the hierarchical feature representation from the original data automatically, deep-learning methods have made breakthroughs in the field of computer vision [24], [25]. Some typical deep-learning architectures have been also applied to HSI classification and made great progress, such as convolutional neural networks (CNNs) [26]–[28], recurrent neural networks (RNNs) [29], deep belief networks (DBNs) [30], and stacked autoencoder (SAE) [31]. Due to the native ability in extracting spatial and spectral features, CNNs are

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Qichao Liu and Jingxiang Yang are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: qc.l@qq.com; yang123jx@njust.edu.cn;)

Liang Xiao is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China, and also with the Key Lab of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: xiaoliang@mail.njust.edu.cn).

Jonathan Cheung-Wai Chan is with the Department of Electronics and Informatics, Vrije Universiteit Brussel, 1050 Brussels, Belgium (e-mail: jcheungw@etrovub.be).

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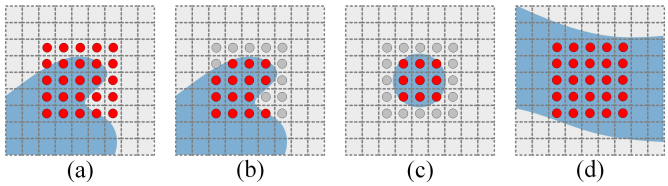

Fig. 1. Illustration of sampling shapes in 5×5 regular and content-guided convolutions, where blue denotes homogeneous pixels, while white denotes other classes of pixels. (a) Regular sampling grid (red points) of regular convolution. (b)–(d) Shape-adaptive sampling grid over the edge, isolated point, and flat region, respectively (red points possess strong weights, while gray points possess small or zero weights).

drawing enormous attention in HSI classification. Different structures of CNN models have been investigated, such as the 1-D-CNN [32], 2-D-CNN [26], and 3-D-CNN [33]. For combining the advantages of different network structures, researchers have also proposed many hybrid networks, such as the two-channel CNN (TC-CNN) [34] containing 1-D- and 2-D-CNNs, hybrid spectral network (HybridSN) [35] containing 2-D- and 3-D-CNNs, and the multichannel CNN (MC-CNN) [36] containing 1-D-, 2-D-, and 3-D-CNNs. High depth is often necessary for these networks to achieve satisfactory performance. It is inevitable that a deeper network contains more parameters and requires more training samples. However, in most cases, there are only limited labeled samples in HSI classification, which leads to overfitting and hinders the application of deeper networks. To deal with the small-sample problem, Yang *et al.* [34] proposed to use transfer learning to supply spectral–spatial information from other data sets to target HSI. Zhong *et al.* [37] and Wang *et al.* [38] used the residual connection [39] and the dense connection [40], respectively, to enhance the robustness of deep networks to overfitting. Another extension of CNNs is the usage of separable filtering learning [41], [42] or deep learning with separable convolutions [43]. For instance, a novel deep convolutional network architecture inspired by Inception is proposed in [43], where regular 2-D convolutions have been replaced with depthwise separable convolutions. Due to the separable convolutions, the learning procedure is more efficient.

However, most of the deep-learning-based HSI classification methods tend to oversmooth classification maps and fail to classify the pixels near cross-classes edges. One weakness is the fixed shape of the regular convolution kernels. Specifically, in a local region of HSI, the related neighboring pixels (i.e., the same land-cover class with the center pixel) can provide useful information, while the irrelevant neighboring pixels (i.e., different land-cover classes with the center pixel) provide unexpected miscellaneous information. As shown in Fig. 1(a), due to the fixed kernel shape (usually rectangular), the regular convolution would sample pixels belonging to different classes, particularly near the cross-classes edges; thus, the feature attributes of other classes will be included. If the kernel shape could change adaptively, it would be possible to filter out the irrelevant pixels and preserve only the relevant pixels during feature extraction, as shown in Fig. 1(b)–(d). In the computer vision field, researchers have proposed some convolutions with irregular kernels according to semantic information. For example, Dai *et al.* [44] proposed a deformable convolutional network (DCN) that added additional offsets to the sampling locations of convolution kernels, which explicitly enabled convolution kernels to sample feature maps at arbitrary locations instead of a fixed one. Jaderberg *et al.* [45] proposed to adjust the kernel shape in an implicit fashion by translating, rotating, and scaling the feature maps; thus, the sampling mode of convolution kernel can be adaptive to image content. These irregular CNNs with deformation ability could effectively characterize the semantic structures of images. However, how to effectively characterize the spectral–spatial structures of HSIs in the context of CNN-based HSI classification still needs more study.

In this article, we propose a novel content-guided convolution that could adaptively adjust its kernel shape according to the content of HSIs. The content of HSI can be reflected by the spatial distribution of land covers, which is described by a latent guide map (LGM) automatically learned from training samples. With the proposed content-guided convolution, we propose a bandwise content-guided convolution to reduce the parameters of the deep network and enhance its robustness to overfitting. Based on the bandwise content-guided convolution, a guided feature extraction unit (GFEU) to extract spectral–spatial features jointly from HSIs is built. Multiple GFEUs are densely connected to form the final content-guided CNN (CGCNN) for HSI classification, as shown in Fig. 2.

The main contributions of this article are as follows.

1) We propose an LGM learning module to characterize the spatial distribution of HSIs. The LGM is automatically learned, which endows the LGM with great generalization and adaptability.
2) We propose a novel content-guided convolution with content-adaptive kernel shape, which could adjust its kernel shape according to the LGM and explicitly utilize the spatial distribution of HSI. We also propose the bandwise content-guided convolution with fewer parameters, and it could enhance the robustness of overfitting.
3) By using content-guided convolutions, we construct a novel GFEU that can effectively model diverse local regions and extract spectral–spatial features jointly from them. Multiple GFEUs are densely connected to form the final CGCNN for HSI classification; such network architecture shows promising performance.

The rest of this article is organized as follows. The proposed CGCNN is described in Section II. The experiments and analysis are shown in Section III. The conclusion is discussed in Section IV.

II. Content-Guided Convolutional Neural Network

Fig. 2 shows the proposed CGCNN, which is an end-to-end HSI classification framework. The input $\mathbf{X} \in \mathbb{R}^{H \times W \times B}$ denotes the original HSI, and the output $\mathbf{Y} \in \mathbb{R}^{H \times W \times C}$ denotes the class probability of each pixel, where H, W, B, and C denote the height, width, number of bands, and number of classes, respectively. Compared with conventional deep-learning-based

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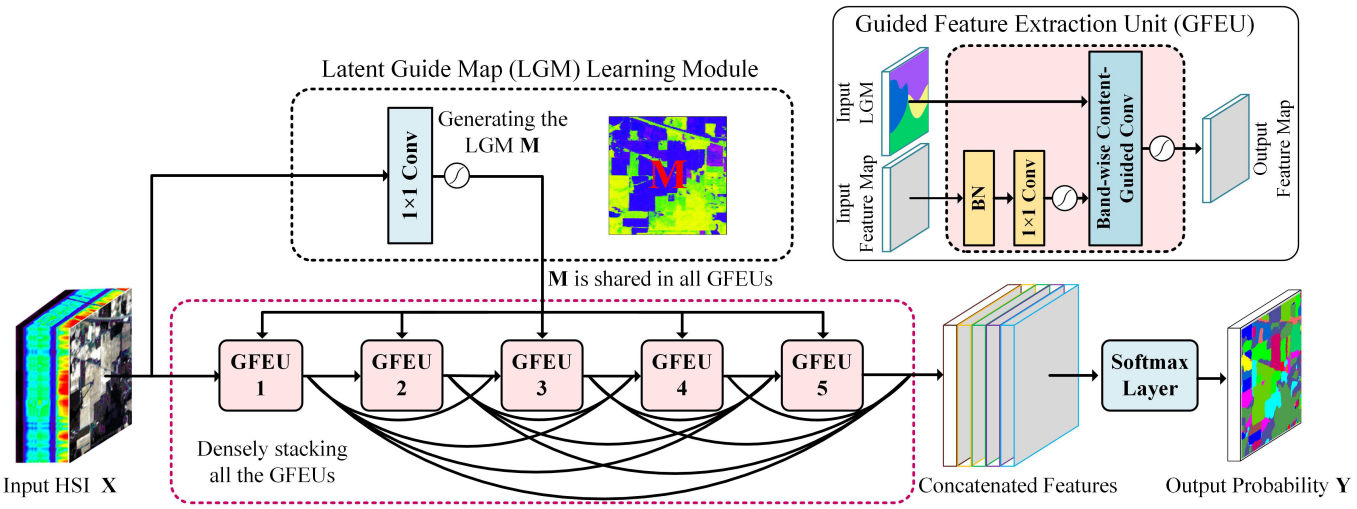

Fig. 2. Proposed CGCNN architecture where **X**, **Y**, and **M** denote the input, output, and LGM, respectively.

HSI classification framework, CGCNN is different in three aspects: 1) the use of content-guided convolutions; 2) LGM learning module; and 3) GFEUs are the basic building blocks and densely stacked. In the following, we will describe in detail each building block of CGCNN.

A. Latent Guide Map Learning Module

In this section, we describe, in detail, the LGM, including the module architecture and learning method. To describe the content of HSI that can then be used to guide the kernel shape of CGCNN, we propose to learn a nonlinear transformation to extract spatial patterns of different land covers. Let $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ be the HSI and $\mathbf{M} \in \mathbb{R}^{H\times W\times M}$ be the output LGM, where M is the number of channels of the LGM. For each spatial location $\mathbf{p}_0 = (x, y)$ on the LGM, we have

$$\mathbf{M}_j(\mathbf{p}_0) = f\left(\mathbf{W}_j \cdot \mathbf{X}(\mathbf{p}_0) + b_j\right) \tag{1}$$

where $\mathbf{W}_j$ denotes the jth 1-D kernel along the spectral dimension of HSI, b_j denotes the jth bias, $\mathbf{X}(\mathbf{p}_0)$ denotes the pixel of HSI $\mathbf{X}$ at spatial location $\mathbf{p}_0$, $\mathbf{M}_j(\mathbf{p}_0)$ is the output value of the jth channel of LGM at location $\mathbf{p}_0$, the operator $\cdot$ denotes inner product, and f denotes the sigmoid function. This transformation is applied on each pixel, which is equivalent in implementation to a standard 2-D convolution layer with 1×1 size kernels, as shown in the LGM learning module in Fig. 2.

Note that our proposed LGM learning module is a generalization of PCA that is to find a transformation to reduce the high dimensionality of spectra and preserve most of the spectral information in the first several PCs. In [28], PCA was used to extract the spatial structure of HSIs. Since PCA is an unsupervised method with linear transformation, its discriminative ability may be weaker. The proposed LGM learning module is supervised and nonlinear and can be trained with the network.

The LGM characterizes the spatial correlation. Specifically, in a local region of HSI, the pixels with similar spectra will have similar values in the LGM, while the pixels with dissimilar spectra will have very different values. This revealed the spatial structure of HSI can guide the classification near edges or class boundaries. In Section III, we will compare the LGM learning module and PCA in detail.

B. Content-Guided Convolution

In this section, we introduce the proposed content-guided convolution, of which the kernel shape can be steered by the LGM.

1) Regular Convolution: The regular 2-D convolution shown in Fig. 3(a) has two steps: sampling the input feature map $\mathbf{x}$ with a regular grid $\mathcal{R}$ and then weighting the sampled values by $\mathbf{k}$ and summarizing them to a feature. The grid $\mathcal{R}$ describes a typical sampling grid, for instance

$$\mathcal{R} = \{(-1,-1),(-1,0),\ldots,(0,1),(1,1)\} \tag{2}$$

describes a regular 3×3 sampling grid. In the regular 2-D convolution, for each spatial location $\mathbf{p}_0 = (x, y)$ on the output feature map $\mathbf{y}$, we have

$$\mathbf{y}_j(\mathbf{p}_0) = \sum_{\mathbf{p}_n\in\mathcal{R}} \mathbf{k}_j(\mathbf{p}_n) \cdot \mathbf{x}(\mathbf{p}_0 + \mathbf{p}_n) \tag{3}$$

where $\mathbf{p}_n$ enumerates the locations in $\mathcal{R}$, $\mathbf{k}_j(\mathbf{p}_n)$ denotes the vector of the jth regular kernel at location $\mathbf{p}_n$, $\mathbf{x}(\mathbf{p}_0 + \mathbf{p}_n)$ denotes the pixel of input feature map at spatial location $\mathbf{p}_0 + \mathbf{p}_n$, and $\mathbf{y}_j(\mathbf{p}_0)$ is the output value of the jth output feature channel at location $\mathbf{p}_0$. Note that for the convenience of description, the bias and activation function are not considered here.

2) Content-Guided Convolution: In the content-guided convolution shown in Fig. 3(b), the shape of kernel $\mathbf{G}$ (called guided kernel) is steered by an anisotropic kernel that is calculated from the local region of LGM $\mathbf{M}$ at the current sampling location $\mathbf{p}_0$ with the same sampling grid. Then, (3) becomes

$$\mathbf{y}_j(\mathbf{p}_0) = \sum_{\mathbf{p}_n\in\mathcal{R}} \mathbf{G}_j(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M}) \cdot \mathbf{x}(\mathbf{p}_0 + \mathbf{p}_n) \tag{4}$$

where $\mathbf{G}_j(\mathbf{p}_0, \cdot; \mathbf{M})$ denotes the jth guided kernel at current sampling location $\mathbf{p}_0$, and $\mathbf{y}_j$ denotes the jth channel of the

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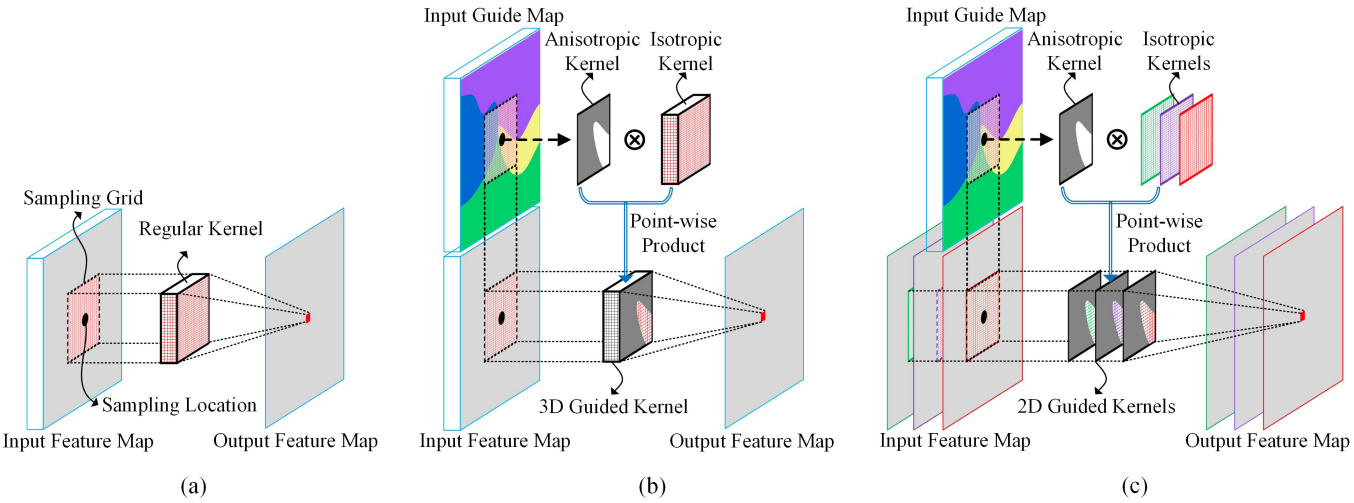

Fig. 3. Different convolution structures. (a) Regular convolution. (b) Content-guided convolution. (c) Bandwise content-guided convolution.

output feature map. The guided kernel can be separated into two kernels

$$\mathbf{G}_j(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M}) = g(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M})\mathbf{k}_j(\mathbf{p}_n) \tag{5}$$

where $\mathbf{k}_j$, which is the same as the regular kernel in (3), denotes the isotropic kernel in the jth guided kernel, and $g(\mathbf{p}_0, \cdot; \mathbf{M})$ denotes the anisotropic kernel calculated by

$$g(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M}) = \exp\left(-\frac{\|\mathbf{M}(\mathbf{p}_0 + \mathbf{p}_n) - \mathbf{M}(\mathbf{p}_0)\|_2^2}{\sigma^2}\right) \tag{6}$$

where $\mathbf{M}$ is the LGM, σ is the sensitivity value for adjusting the anisotropy degree, $\|\cdot\|_2$ represents the L2-norm used to compute the Euclidean distance between two vectors, and $\exp(\cdot)$ denotes the exponential function. The anisotropic kernel g is calculated from the LGM at current sampling location $\mathbf{p}_0$, i.e., the anisotropic kernels at different sampling locations are different. The isotropic kernel $\mathbf{k}_j$ does not depend on the sampling location, i.e., $\mathbf{k}_j$ is shared in the convolution procedure. Note that g is two dimensional, while $\mathbf{k}_j$ is three dimensional, each slice of $\mathbf{k}_j$ along spectral dimension needs to be multiplied by g point by point.

The sensitivity value σ in (6) is a trainable parameter in the network, which is updated by the backpropagation (BP) [46] algorithm in the training process. Specifically, in the content-guided convolution layer, the error is first propagated to the guided kernel $\mathbf{G}_j$ and then propagated to the anisotropic kernel g and sensitivity value σ by the chain rule. We can calculate the differential equation of σ by

$$\begin{aligned}\frac{\partial \mathbf{G}_j(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M})}{\partial \sigma} &= \frac{\partial \mathbf{G}_j(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M})}{\partial g(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M})} \frac{\partial g(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M})}{\partial \sigma} \\ &= \mathbf{k}_j(\mathbf{p}_n)\frac{2\theta}{\sigma^3}\exp\left(-\frac{\theta}{\sigma^2}\right)\end{aligned} \tag{7}$$

where $\theta = \|\mathbf{M}(\mathbf{p}_0 + \mathbf{p}_n) - \mathbf{M}(\mathbf{p}_0)\|_2^2$. Then, σ is updated by

$$\sigma \leftarrow \sigma - r \times \frac{\partial \mathcal{L}}{\partial \mathbf{G}_j(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M})} \frac{\partial \mathbf{G}_j(\mathbf{p}_0, \mathbf{p}_n; \mathbf{M})}{\partial \sigma} \tag{8}$$

where $\mathcal{L}$ denotes the loss function of network and r is the learning rate. Note that the update procedure of σ is a gradient descent method. In practice, we use the automatic differentiation tool of Tensorflow [47] to calculate the differential equations of trainable parameters and update them.

3) Bandwise Content-Guided Convolution: In general, available training samples in HSI classification are limited, while the amount of parameters in the deep network is huge, reducing the parameters of the network could enhance the resistance to overfitting. Inspired by the depthwise separable convolution in [43], which could extract spatial features with fewer parameters compared with the regular convolution, we also propose the bandwise content-guided convolution. Specifically, in the bandwise version of content-guided convolution shown in Fig. 3(c), the input feature map is sliced into 2-D bands along a spectral dimension, of which each band is convoluted with a unique 2-D guided kernel and output a 2-D feature band by using (4).

Note that in the bandwise content-guided convolution, the guided kernel $\mathbf{G}_j$ is two dimensional. The numbers of bands (or channels) of input and output feature maps are the same and are equal to the number of 2-D guided kernels. Because the bandwise version of content-guided convolution can only extract spatial features of each input band instead of spectral features, to exploit spectral information along with spatial information of HSI, the bandwise convolution layer needs to be combined with a 1×1 size 2-D convolution layer in general. Assuming that the input feature map has n bands, the spatial size of convolution kernel is $s \times s$, and then, outputting each feature band requires $n \times s^2$ parameters in the original content-guided convolution. However, in the bandwise convolution combined with 1×1 convolution (1×1 convolution is in the front of bandwise convolution, as shown in the GFEU in Fig. 2), only $n + s^2$ parameters are required. By using the bandwise version of content-guided convolution, the parameters of network can be reduced greatly, which could reduce the risk of overfitting in the case of limited samples.

C. Guided Feature Extraction Unit

Features extracted from the spatial and spectral domains are beneficial for HSI classification. Currently, researchers have

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proposed many multibranch CNN architectures, in which each branch extracted a specific type of features (e.g., spectral or spatial features) independently [27], [28], [36]. However, recent works indicate that extracting the spectral and spatial features in a common CNN branch can reduce parameters and avoid overfitting in the cases of limited samples [37], [38]. In this article, based on the bandwise content-guided convolution, we propose a hidden unit containing a series of layers to extract spectral and spatial features jointly, and we name it GFEU.

As shown in Fig. 2, a GFEU has two inputs: the feature map produced from the previous layer and the LGM computed by (1). The GFEU takes the feature map in the previous layer as input, and batch normalization (BN) [48] is adopted to reduce the covariance drift. Let $\mathbf{I}_l \in \mathbb{R}^{H\times W\times U}$ be the input feature map of the lth GFEU, where H, W, and U denote the height, width, and bands of the input feature map, respectively, and then, the normalized input data $\tilde{\mathbf{I}}_l \in \mathbb{R}^{H\times W\times U}$ can be calculated by

$$\tilde{\mathbf{I}}_l = \frac{\mathbf{I}_l - E(\mathbf{I}_l)}{\mathrm{Var}(\mathbf{I}_l)} \tag{9}$$

where $E(\cdot)$ and $\mathrm{Var}(\cdot)$ represent the expectation and variance function, respectively. After normalization, we use a 1×1 size convolution to apply a nonlinear transformation on each pixel in $\tilde{\mathbf{I}}_l$ to extract spectral features with reduced dimensionality. For each location $\mathbf{p}_0$ on the output spectral feature map $\mathbf{I}_l \in \mathbb{R}^{H\times W\times V}$, where V denotes the number of the output bands, we have

$$\mathbf{T}_{l,j}(\mathbf{p}_0) = f(\mathbf{k}_{l,j}\cdot\tilde{\mathbf{I}}_l(\mathbf{p}_0) + b_{l,j}) \tag{10}$$

where $\mathbf{k}_{l,j}$ and $b_{l,j}$ are the jth 1-D kernel and bias in the lth GFEU, respectively, and $\mathbf{T}_{l,j}$ represents the jth channel of the output feature map. The 1×1 convolution is able to extract the spectral features of each pixel, and then, a bandwise content-guided convolution is conducted on the spectral feature map $\mathbf{T}_l$ to exploit the spatial information from the neighborhood of each pixel. For each location $\mathbf{p}_0$ on the output spectral–spatial feature map $\mathbf{O}_l \in \mathbb{R}^{H\times W\times V}$, we have

$$\mathbf{O}_{l,j}(\mathbf{p}_0) = f\left(\sum_{\mathbf{p}_n\in\mathcal{R}} \mathbf{G}_{l,j}(\mathbf{p}_0,\mathbf{p}_n;\mathbf{M})\cdot\mathbf{T}_{l,j}(\mathbf{p}_0+\mathbf{p}_n) + p_{l,j}\right) \tag{11}$$

where $\mathbf{G}_{l,j}$ and $p_{l,j}$ are the jth guided kernel and bias in the lth GFEU, respectively, $\mathbf{T}_{l,j}$ and $\mathbf{O}_{l,j}$ denote the jth channel of the input and output feature map, respectively, and $\mathcal{R}$ defines the sampling grid.

D. Architecture of CGCNN

Gradient vanishing is a challenge for deep neural networks. In [40], DenseNet has been proposed to address this problem. Different layers in DenseNet are connected densely, so the gradient can be propagated directly among different layers, which helps to avoid gradient vanishing and overfitting. Inspired by DenseNet, we propose to densely connect different GFEUs in our network. Let $\mathbf{I}_l$ and $\mathbf{O}_l$ be the input and output of the lth GFEU, respectively, and then, $\mathbf{I}_l$ can be obtained by

$$\mathbf{I}_l = \left[\mathbf{O}_1, \mathbf{O}_2, \ldots, \mathbf{O}_{l-1}\right] \tag{12}$$

where $[\ldots]$ denotes the concatenation of different feature maps along with spectral dimension.

One of the advantages of the dense connection is that the features produced by previous layers can be reused, which makes the network more capable for feature representation. Assuming that the network is composed of L ($1 \leqslant L$) GFEUs, the final feature maps $\mathbf{I}$ fed into the softmax classifier are obtained by

$$\mathbf{I} = [\mathbf{O}_1, \mathbf{O}_2, \ldots, \mathbf{O}_L]. \tag{13}$$

A softmax classifier is used to calculate the probabilities of each class at each pixel. Let $\mathbf{Y}$ denote the output probability map; for each location $\mathbf{p}_0$ on the output probability map $\mathbf{Y}$, we have

$$\mathbf{Y}(\mathbf{p}_0) = \frac{1}{\sum_{m=1}^{C} e^{\mathbf{k}_m\cdot\mathbf{I}(\mathbf{p}_0)+b_m}} \begin{bmatrix} e^{\mathbf{k}_1\cdot\mathbf{I}(\mathbf{p}_0)+b_1} \\ e^{\mathbf{k}_2\cdot\mathbf{I}(\mathbf{p}_0)+b_2} \\ \vdots \\ e^{\mathbf{k}_C\cdot\mathbf{I}(\mathbf{p}_0)+b_C} \end{bmatrix} \tag{14}$$

where $\mathbf{I}$ is the concatenated feature maps, $\mathbf{k}_m$ and $b_m (1 \leqslant m \leqslant C)$ denote the mth 1-D kernel and bias, C is the number of classes, and $\mathbf{Y}(\mathbf{p}_0)$ denotes the class probability of the pixel at location $\mathbf{p}_0$.

In order to deal with the problem of category imbalance in training samples, a set of weights is proposed to adjust the class loss, i.e., classes with fewer samples will be considered preferentially in training process. Specifically, let N_c ($1 \leqslant c \leqslant C$) denote the number of supervised pixels of the cth class. In order to increase the preference of the classes with fewer samples, we propose to multiply the loss of the cth class with the weight $1/N_c$. Let $\mathcal{D} = \{(\mathbf{X}(\mathbf{p}_t), \mathbf{L}(\mathbf{p}_t))\}$ represent the set composed of all the training samples in the HSI, where $\mathbf{L}(\mathbf{p}_t)$ denotes the label vector of the training sample $\mathbf{X}(\mathbf{p}_t)$, $\mathbf{p}_t$ ($t = 1, 2, \ldots, N$) denotes the spatial location of training sample, and N is the number of training samples, and then, the class-weighted cross entropy loss function is

$$\mathcal{L} = -\sum_{t=1}^{N}\sum_{c=1}^{C} \frac{1}{N_c}\mathbf{L}_c(\mathbf{p}_t)\log(\mathbf{Y}_c(\mathbf{p}_t)) \tag{15}$$

where $\mathbf{L}_c(\mathbf{p}_t)$ denotes the cth element of the label vector $\mathbf{L}(\mathbf{p}_t)$, and $\mathbf{Y}_c(\mathbf{p}_t)$ denotes the probability of the cth class of the pixel $\mathbf{X}(\mathbf{p}_t)$.

III. Experiments

We have used four widely used HSI data sets to validate the proposed CGCNN: synthetic data set (SD) [20], Indian Pines (IP), Kennedy Space Center (KSC), and University of Pavia (UP). Six state-of-the-art deep-learning-based HSI classification methods are compared with our method: 2-D-CNN [26], TC-CNN [34], 3-D-CNN [33], MC-CNN [36], SSRN [37], and HybridSN [35]. Overall accuracy (OA), average accuracy (AA), and kappa statistics are used to evaluate the classification performance.

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TABLE I
CATEGORY INFORMATION OF THE SD

No.	Class	Train.	Val.	Test.
1	Class-1	75	75	7391
2	Class-2	83	83	8172
3	Class-3	23	23	2272
4	Class-4	105	105	10313
5	Class-5	112	112	11056
Total		398	398	39204

A. Configuration for Hyperparameters

CGCNN includes an LGM learning module and five GFEUs. Specifically, the number of output channels of the LGM learning module is 3, and the number of channels of the output feature map is 128 for the first GFEU and 32 for the remaining GEFUs. In all GFEUs, the size of the guided kernel is set to 5×5. The initial value of the sensitivity parameter σ in (6) is set to 1. Finally, we use the Adam optimizer [49] to train our network, in which different parameters have different learning rates. Specifically, the learning rate of σ is 0.005, and the learning rate of the remaining parameters (e.g., convolution kernels) is 0.0005. The number of training iterations is 800. Our code[1] is completed with Python 3.6 and Tensorflow-1.12. The experiments are conducted using the following environment: CPU: i7-8700K, GPU: GTX-1080Ti, and memory: 32 GB.

B. Data Set

The first data set SD is obtained from [20], which consists of 162 spectral bands with wavelength ranging from 0.4 to 2.5 μm. The spatial dimension is 200×200. A total of 40 000 samples of five classes are in this data set. We randomly chose 1%, 1%, and 98% samples from each class as the training, validation, and testing data sets. The detailed numbers of samples of each class are listed in Table I.

The second data set IP was captured over the agricultural regions in the IP test site by the Airborne Visible-Infrared Imaging Spectrometer (AVIRIS) sensor. This HSI consists of 220 bands with an image size of 145×145. After removing 20 bands of water vapor absorption and low signal-to-noise ratio (band numbers: 104–108, 150–163, and 220), there are remaining 200 bands used for the experiment. A total of 10 366 samples of 16 terrain classes were collected in this area. We randomly chose 10%, 1%, and 89% samples from each class as the training, validation, and testing data sets, as listed in Table II.

The third data set is the KSC acquired by the AVIRIS sensor over KSC, Florida, in 1996. Its spatial resolution is 18 m, and its image size is 512×614. After removing water absorption and low signal-to-noise-ratio bands, the data set includes 176 bands. There are 13 classes of land covers in this data set. We randomly chose 5%, 1%, and 94% samples from each class as the training, validation, and testing data sets, as listed in Table III.

The fourth data set is the UP that was captured over the UP by the Reflective Optics System Imaging Spectrometer (ROSIS-3) sensor. This HSI consists of 115 bands in the range of 0.43–0.86 μm with the spatial resolution of 20 m, and the image size is 610×340. After removing low signal-to-noise ratio and water absorption bands, the remaining 103 bands with nine classes of land covers were used. We randomly chose 1%, 1%, and 98% samples from each class as the training, validation, and testing data sets, as listed in Table IV.

[1]Code is available at https://github.com/qichaoliu/Content_Guided_CNN

TABLE II
CATEGORY INFORMATION OF THE INDIANA PINES DATA SET

No.	Class	Train.	Val.	Test.
1	Alfalfa	5	1	48
2	Corn-notill	143	14	1277
3	Corn-mintill	83	8	743
4	Corn	23	2	209
5	Pasture	49	4	444
6	Trees/Grass	74	7	666
7	Pasture-mowed	2	1	23
8	Hay-windrowed	48	4	437
9	Oats	2	1	17
10	Soybeans-notill	96	9	863
11	Soybean-mintill	246	24	2198
12	Soybean-cleantill	61	6	547
13	Wheat	21	2	189
14	Woods	129	12	1153
15	Building-Grass	38	3	339
16	Stone-steelTowers	9	1	85
	Total	1029	99	9238

TABLE III
CATEGORY INFORMATION OF THE KSC DATA SET

No.	Class	Train.	Val.	Test.
1	Scrub	18	4	325
2	Willow swamp	13	3	227
3	CP hammock	13	3	240
4	Slash pine	13	3	236
5	Oak/broadleaf	9	2	150
6	Hardwood	12	3	214
7	Swamp	6	2	97
8	Graminoid marsh	20	4	366
9	Spartina marsh	26	6	488
10	Cattail marsh	21	5	378
11	Salt marsh	21	5	393
12	Mud flats	26	6	471
13	Water	47	10	870
	Total	245	56	4455

TABLE IV
CATEGORY INFORMATION OF THE UP DATA SET

No.	Class	Train.	Val.	Test.
1	Asphalt	67	67	6497
2	Meadows	187	187	18275
3	Gravel	21	21	2057
4	Trees/Grass	31	31	3002
5	Metalsheets	14	14	1317
6	Baresoil	51	51	4927
7	Bitumen	14	14	1302
8	Bricks	37	37	3608
9	Shadows	10	10	927
	Total	432	432	41912

C. Comparison of Classification Performance

We compare the performance of CGCNN with six deep-learning-based methods on four HSI data sets. Each experiment is repeated ten times, and the mean and the standard

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TABLE V
INDIVIDUAL CLASS, OA, AA, AND KAPPA STATISTICS OF ALL METHODS ON THE SD

Class	2D-CNN [26]	TC-CNN [34]	3D-CNN [33]	MC-CNN [36]	SSRN [37]	HybridSN [35]	CGCNN
1	94.57±0.68	87.95±2.36	98.92±1.45	97.55±0.49	88.78±1.47	86.64±1.17	**99.84±0.31**
2	96.06±1.46	88.80±2.48	97.78±1.54	96.37±1.10	92.07±1.28	87.39±0.94	**99.91±0.18**
3	92.68±3.24	90.90±1.40	98.10±3.22	**98.17±0.32**	90.58±2.42	82.22±4.77	94.89±1.09
4	95.14±0.60	91.18±1.26	**99.05±0.78**	97.09±1.16	90.41±2.06	87.20±1.58	98.91±0.26
5	96.81±1.42	91.87±1.55	99.56±0.25	99.11±0.26	93.64±1.49	89.47±1.44	**99.74±0.28**
OA(%)	95.55±0.47	90.17±0.42	98.85±0.40	97.63±0.25	91.29±0.45	87.45±0.47	**99.29±0.29**
AA(%)	95.05±0.79	90.14±0.52	**98.68±0.49**	97.66±0.24	91.10±0.73	86.58±0.98	98.66±0.40
Kappa(×100)	94.21±0.61	87.20±0.54	98.50±0.52	96.92±0.33	88.66±0.58	83.67±0.61	**99.08±0.38**

TABLE VI
INDIVIDUAL CLASS, OA, AA, AND KAPPA STATISTICS OF ALL METHODS ON THE IP DATA SET

Class	2D-CNN [26]	TC-CNN [34]	3D-CNN [33]	MC-CNN [36]	SSRN [37]	HybridSN [35]	CGCNN
1	**100±0**	95.65±8.69	99.41±1.17	98.18±3.63	99.42±1.14	97.16±2.73	**100±0**
2	94.11±0.68	95.84±0.92	96.07±2.27	96.25±0.80	98.40±0.96	97.15±0.56	**99.03±0.29**
3	93.69±1.52	97.31±1.07	94.65±1.75	96.84±1.02	97.13±2.29	98.25±0.65	**99.49±0.11**
4	95.40±4.35	96.85±1.72	97.65±1.97	97.16±2.46	96.71±4.20	97.90±2.57	**100±0**
5	96.87±1.10	97.88±1.13	98.76±1.09	99.03±0.17	97.20±1.30	98.49±0.96	**99.77±0.14**
6	98.35±1.29	98.51±1.35	98.00±1.34	98.61±1.29	99.01±0.89	98.92±0.38	**99.92±0.08**
7	**100±0**	98.82±2.35	98.82±2.35	98.03±2.41	80±40	**100±0**	99.24±1.85
8	96.58±1.02	98.79±1.54	99.09±0.97	99.45±0.68	99.45±0.61	99.67±0.31	**99.96±0.09**
9	**100±0**	92.51±1.03	95.00±6.12	97.64±4.70	0±0	92.38±5.25	95.26±4.38
10	94.27±2.43	95.43±1.92	96.18±0.91	95.21±1.72	96.38±3.06	**98.74±0.81**	97.11±0.79
11	95.81±1.23	97.51±0.47	96.08±1.16	96.68±0.92	98.02±0.90	99.16±0.26	**99.23±0.19**
12	93.74±3.45	95.12±1.31	97.02±1.36	96.04±1.43	95.42±4.10	97.47±1.17	**98.93±0.79**
13	99.68±0.63	98.95±0.32	99.78±0.26	99.58±0.51	98.95±1.27	98.02±1.64	**99.90±0.22**
14	98.39±0.93	99.01±0.77	98.82±0.59	99.30±0.36	99.48±0.28	99.32±0.44	**99.60±0.45**
15	95.67±1.98	94.83±2.38	94.72±2.13	95.71±2.18	96.76±1.89	97.64±1.58	**98.95±0.78**
16	94.48±4.08	96.16±2.55	95.48±2.79	95.48±2.20	97.05±2.63	91.02±4.07	**98.39±0.59**
OA(%)	95.79±0.44	97.14±0.31	96.84±0.84	97.19±0.45	97.86±0.41	98.44±0.16	**99.17±0.15**
AA(%)	96.69±0.40	96.82±1.05	97.22±0.57	97.45±0.53	90.59±2.34	97.58±0.82	**99.05±0.35**
Kappa(×100)	95.20±0.50	96.74±0.36	96.40±0.95	96.80±0.52	97.56±0.47	98.23±0.18	**99.06±0.17**

TABLE VII
INDIVIDUAL CLASS, OA, AA, AND KAPPA STATISTICS OF ALL METHODS ON THE KSC DATA SET

Class	2D-CNN [26]	TC-CNN [34]	3D-CNN [33]	MC-CNN [36]	SSRN [37]	HybridSN [35]	CGCNN
1	97.65±1.05	98.54±0.94	98.58±0.17	98.97±0.96	98.84±0.94	98.47±0.51	**99.23±0.38**
2	88.78±4.42	90.17±3.28	82.31±5.79	89.13±7.24	97.07±5.63	93.75±5.50	**98.18±1.78**
3	64.38±11.24	74.71±3.36	73.55±8.35	78.51±5.63	97.13±1.27	89.08±2.36	**98.65±0.16**
4	67.29±7.56	78.33±11.70	60.33±8.67	76.57±8.41	89.77±4.52	89.93±3.01	**93.17±2.92**
5	65.42±14.22	63.94±10.20	64.64±11.99	61.00±7.30	87.78±12.10	90.48±6.70	**98.66±0.36**
6	79.15±4.40	89.88±4.40	79.29±2.96	86.72±1.49	99.32±0.91	92.98±6.09	**100±0**
7	80.14±3.83	87.94±8.17	77.98±11.43	82.79±4.84	93.55±5.53	91.07±6.22	**100±0**
8	88.49±8.07	92.59±5.64	93.37±3.28	92.93±4.68	**98.59±1.23**	93.59±1.90	98.54±1.13
9	90.87±6.95	94.10±2.67	88.06±3.97	88.86±4.86	98.50±1.90	94.12±2.96	**99.62±0.57**
10	99.52±0.58	95.57±5.11	98.36±1.18	99.15±0.45	99.32±1.10	98.36±2.76	**99.73±0.52**
11	99.64±0.44	99.00±1.25	99.69±0.49	99.84±0.20	99.74±0.50	97.69±2.71	**100±0**
12	97.98±1.04	96.34±2.77	89.44±3.08	95.47±2.04	98.16±2.27	99.19±1.19	**99.57±0.35**
13	98.98±1.18	**100±0**	98.56±1.16	99.21±1.34	**100±0**	99.47±0.58	**100±0**
OA(%)	91.06±2.14	93.02±0.99	89.98±1.52	92.44±1.08	97.88±0.80	96.04±1.16	**99.14±0.14**
AA(%)	86.02±2.47	89.32±1.81	84.93±2.46	88.40±1.52	96.75±1.30	94.47±1.94	**98.87±0.26**
Kappa(×100)	90.04±2.39	92.23±1.10	88.84±1.70	91.58±1.20	97.64±0.89	95.59±1.29	**99.09±0.16**

deviation of each index (i.e., OA, AA, and kappa) are reported. The classification results are detailed in Tables V–VIII.

For the SD data set, our method outperforms other compared methods. Due to the serious misclassification at the pixels near cross-classes edges, the 2-D-CNN, TC-CNN, SSRN, and HybridSN obtain oversmoothed classification maps with lower accuracies. It shows that the 2-D structured CNNs are good at modeling the flat regions but fail to characterize other structures, such as isolated points and edge regions. The classification maps obtained by different methods are illustrated in Fig. 4. The results indicate that the proposed method can capture well the spatial structure and patterns of SD, so satisfactory performance has been achieved in the cross-classes regions.

For the IP data set, our method also achieves the best classification result compared with other methods. Because of the small between-classes distance and large within-classes distance in the IP data set, in this case, more neighboring information is necessary to improve the discrimination of different classes. In this experiment, the sensitivity parameter

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TABLE VIII
INDIVIDUAL CLASS, OA, AA, AND KAPPA STATISTICS OF ALL METHODS ON THE UP DATA SET

Class	2D-CNN [26]	TC-CNN [34]	3D-CNN [33]	MC-CNN [36]	SSRN [37]	HybridSN [35]	CGCNN
1	92.72±1.02	89.65±3.13	87.21±3.39	88.65±2.10	**99.66±0.24**	95.13±1.81	98.59±0.72
2	97.14±0.59	97.24±0.86	94.10±2.04	97.37±0.78	98.70±1.02	99.16±0.49	**99.65±0.23**
3	87.91±3.40	74.12±4.36	64.08±6.46	73.78±5.75	93.95±5.15	88.73±4.90	**95.28±2.31**
4	99.35±0.90	97.36±1.34	96.82±1.76	98.54±0.42	**99.72±0.27**	98.18±0.77	97.01±1.96
5	98.92±1.61	98.94±1.23	95.13±4.87	98.91±0.91	99.93±0.08	98.98±0.93	**100±0**
6	97.41±0.77	95.29±0.83	94.07±1.98	93.85±2.24	98.52±2.11	98.66±0.96	**99.45±0.23**
7	91.99±4.80	83.97±3.46	58.80±5.37	84.47±2.91	96.84±2.22	96.64±2.37	**98.00±1.99**
8	88.41±1.07	81.84±2.49	77.11±2.78	82.04±1.68	88.85±5.74	90.69±2.72	**98.55±2.12**
9	99.41±0.59	93.48±2.21	84.19±4.87	96.05±5.62	99.53±0.55	97.21±1.86	**99.95±0.05**
OA(%)	95.35±0.17	93.01±0.73	88.69±1.99	92.89±1.01	97.54±0.58	97.01±0.69	**98.93±0.25**
AA(%)	94.81±0.56	90.21±0.73	83.50±2.14	90.41±1.26	97.30±0.55	95.93±0.87	**98.50±0.40**
Kappa(×100)	93.81±0.23	90.68±0.99	84.86±2.72	90.55±1.34	96.74±0.77	96.02±0.92	**98.59±0.33**

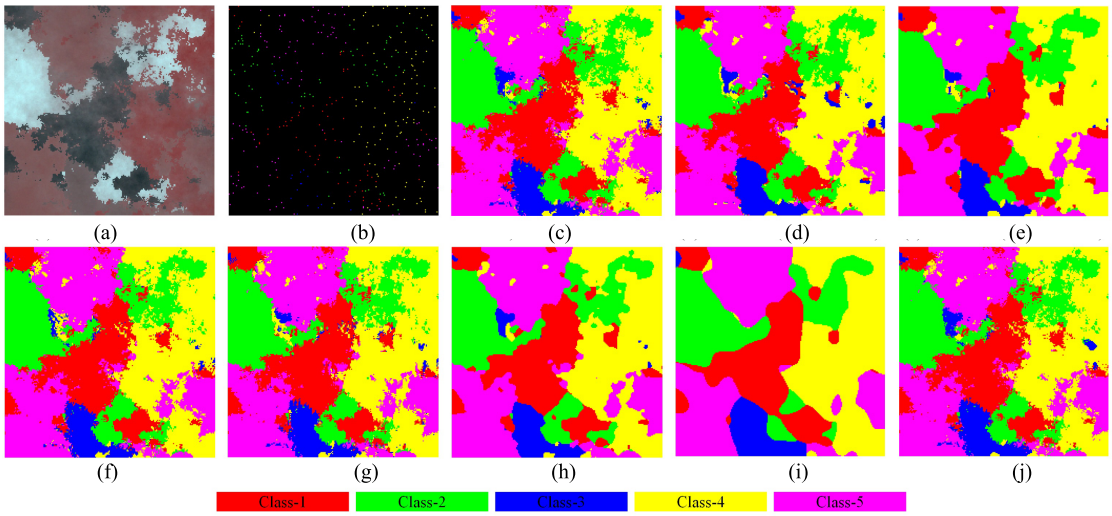

Fig. 4. Classification maps of different methods for the SD. (a) False-color image. (b) Training samples. (c) Ground truth. (d) 2-D-CNN (OA = 95.55%). (e) TC-CNN (OA = 90.17%). (f) 3-D-CNN (OA = 98.85%). (g) MC-CNN (OA = 97.63%). (h) SSRN (OA = 91.29%). (i) HybridSN (OA = 87.45%). (j) CGCNN (OA = 99.29%).

σ in our method is updated to about 0.5, which is larger than that of the SD (the σ of the SD is updated to about 0.2). It indicates that the CGCNN with content-guided convolution encourages piecewise smoothing classification, which results in high consistency with the ground truth. In addition, due to the class-weighted loss function, the AA of CGCNN is higher than all other methods. The classification maps generated by different methods are shown in Fig. 5.

In the experimental results of the KSC and UP, our method still acquires the best accuracies. Because the training sample size of the KSC is relatively small, the reason that our method and SSRN have obtained higher accuracies is attributed to residual connection or dense connection, which is beneficial for preventing overfitting. The classification maps of different methods on the KSC data set are shown in Fig. 6. In addition, the classification map of UP by CGCNN shows sharper edges, while the classification map by HybridSN is obviously oversmoothed; this phenomenon is similar to the experiment of SD. However, the 3-D-CNN, which performs well on the SD, fails to obtain better results on the UP, which indicates that the 3-D convolution is difficult to adapt to different HSI content patterns and lacks robustness in performance. The classification maps of different methods on the UP data set are shown in Fig. 7.

D. Effectiveness of Content-Guided Convolution and LGM Learning Module

In this section, we conduct two ablation experiments to verify the effectiveness of the content-guided convolution and the LGM learning mechanism. Specifically, in the first experiment, we remove the content-guided convolution and replace it with regular convolution while keeps other configuration the same and then compare the classification results of CGCNN with and without the content-guided convolution. Note that the content-guided convolution and regular convolution here are both bandwise. In the second experiment, we replace the learned LGM with the first three PCs of PCA and test its classification performance to show the effectiveness of the LGM learning mechanism. All the experiments in this section adopt the same experimental configuration as in Section III-C.

In the first ablation experiment, as illustrated in Fig. 8, the classification maps obtained by the CGCNN with

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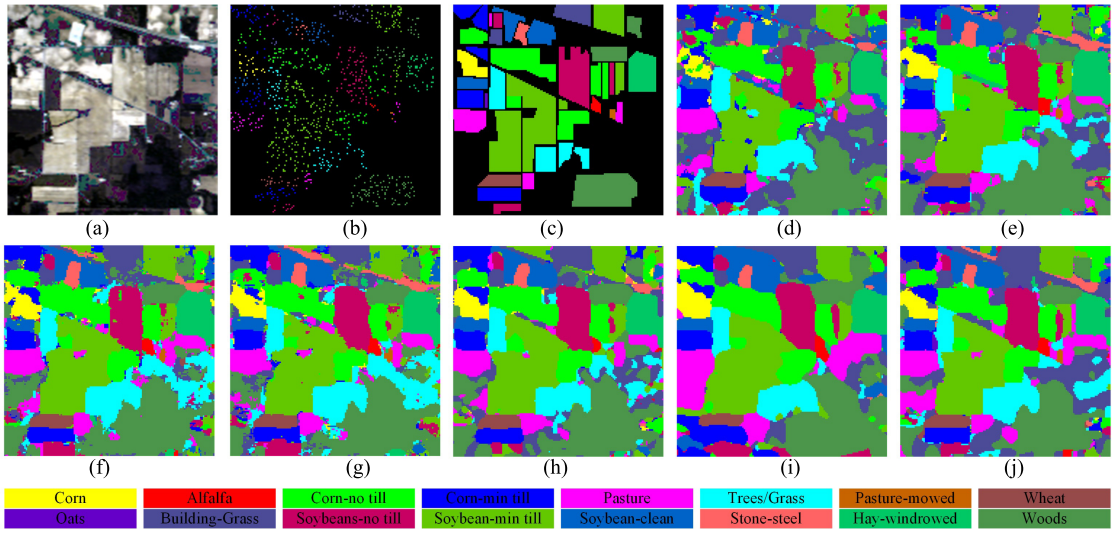

Fig. 5. Classification maps of different methods for the IP data set. (a) False-color image. (b) Training samples. (c) Ground truth. (d) 2-D-CNN (OA = 95.79%). (e) TC-CNN (OA = 97.14%). (f) 3-D-CNN (OA = 96.84%). (g) MC-CNN (OA = 97.19%). (h) SSRN (OA = 97.86%). (i) HybridSN (OA = 98.44%). (j) CGCNN (OA = 99.17%).

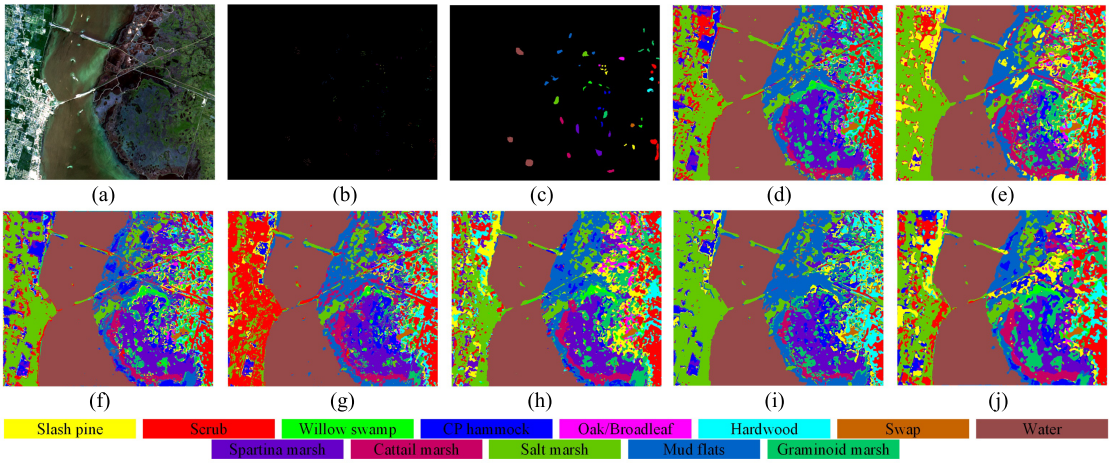

Fig. 6. Classification maps of different methods for the KSC data set. (a) False-color image. (b) Training samples. (c) Ground truth. (d) 2-D-CNN (OA = 91.06%). (e) TC-CNN (OA = 93.02%). (f) 3-D-CNN (OA = 89.98%). (g) MC-CNN (OA = 92.44%). (h) SSRN (OA = 97.88%). (i) HybridSN (OA = 96.04%). (j) CGCNN (OA = 99.14%).

and without content-guided convolution have obvious differences. For instance, in the cross-classes regions [see Fig. 8(b), (c), (e), (i), and (j)], the content-guided convolution is able to preserve the edges or boundaries, while the regular convolution tends to blur them and results in oversmoothed classification maps. In addition, in the isolated points regions [see Fig. 8(a), (d), and (f)–(h)], with the content-guided convolution, the classification maps tend to preserve more isolated points and details. The OA indices of the CGCNN with and without content-guided convolution are reported in Table IX.

We also visualize some representative guided kernels and analyze the effects of them. Due to the content-adaptive property of the CGCNN, the free parameter σ of SD is automatically adjusted to be about 0.2. Hence, the guided kernels over regions with diverse structures of SD possess strong anisotropic attributes, and their shapes are highly consistent with the distribution of land covers, as shown in Fig. 9(a)–(e). In addition, because the IP data set has larger within-classes distance, its sensitivity parameter σ is updated to be about 0.5 automatically, which indicates that the guided kernels of the IP data set use more neighboring information to improve the discrimination of each class while retaining a certain anisotropy ability, as shown in Fig. 9(f)–(j). The above-mentioned results demonstrate that our method can automatically adjust the sensitivity parameter σ to a suitable value and change the shape of the guided kernel according to the HSI content.

In the second ablation experiment, the autolearned LGM is replaced with the first three PCs of PCA. The LGM in our method is automatically learned from the training samples, which makes the LGM more adaptive to different HSIs. The PCs of HSI could also describe the spatial structure and can be regarded as the unsupervised version of LGM. For investigating the advantage of the LGM, we compare the

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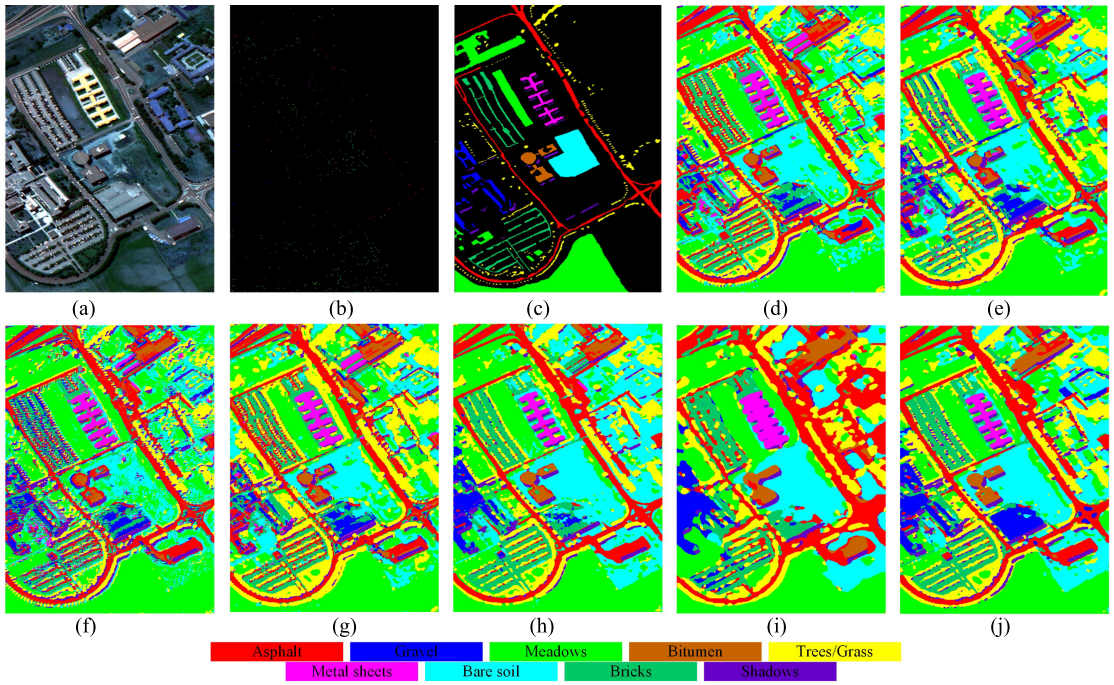

Fig. 7. Classification maps of different methods for the UP data set. (a) False-color image. (b) Training samples. (c) Ground truth. (d) 2-D-CNN (OA = 95.35%). (e) TC-CNN (OA = 93.01%). (f) 3-D-CNN (OA = 88.69%). (g) MC-CNN (OA = 92.89%). (h) SSRN (OA = 97.54%). (i) HybridSN (OA = 97.01%). (j) CGCNN (OA = 98.93%).

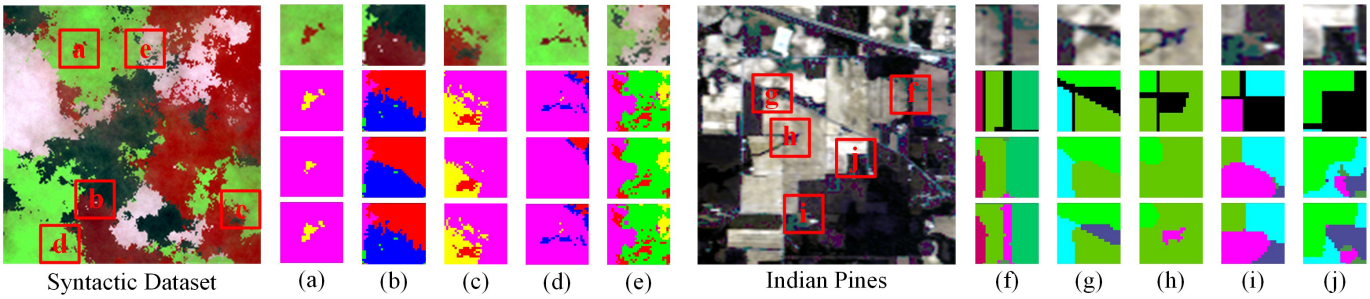

Fig. 8. Comparison of classification details, where the four rows denote the false-color images, the ground truth, the classification maps with regular convolution, and the classification maps with content-guided convolution, respectively. (a)–(e) SD. (f)–(j) IP data set.

TABLE IX
EFFECTIVE ANALYSIS OF CONTENT-GUIDED CONVOLUTION AND LGM LEARNING MODULE (OA%)

Dataset	Content-guided conv. + LGM learning module	Replacing content-guided conv. with regular conv.	Replacing LGM learning module with PCA
SD	99.29	95.59	**99.41**
IP	**99.17**	99.03	99.05
KSC	**99.14**	98.49	98.46
UP	**98.93**	98.68	98.10

classification performance of CGCNNs with LGM learning mechanism and with PCA, and the experimental results are also shown in Table IX. We also show the learned LGMs and the PCs of PCA, as shown in Fig. 10. The results on the IP, KSC, and UP data sets show that the LGM learning is more accurate than PCA, and their corresponding LGMs show clearer boundaries than PCA maps, which means that the LGM could characterize the distribution of land covers better than PCA.

In summary, the first ablation experiment demonstrates that the guided kernels are content adaptive, and they could use related neighboring pixels and discard irrelevant pixels in the local receptive field to boost the classification performance. The second ablation experiment shows that the LGM learning module can extract the spatial structure adaptively and performs better than PCA.

E. Analysis of Hyperparameters in CGCNN

The CGCNN architectures with different hyperparameters (e.g., the size of the guided kernel and the number of GFEUs) may show diverse results. To analyze the effects of CGCNNs with different hyperparameters, we record the OA indices of CGCNNs with different configurations on the four data sets. Specifically, in the first experiment, we set the spatial size of guided kernel to 3×3, 5×5, and 7×7 while fixing the

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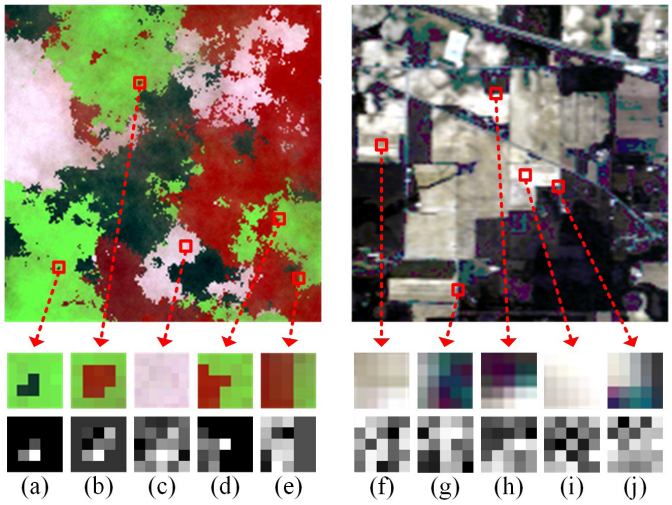

Fig. 9. Visualization of guided kernels in different locations, where the two rows denote the local regions of the false-color images and the guided kernels, respectively. (a)–(e) SD. (f)–(i) IP data set.

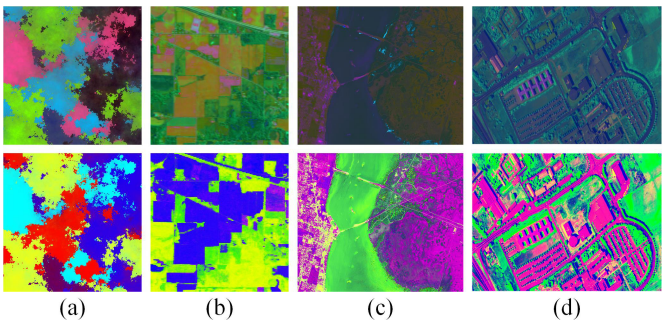

Fig. 10. Comparison of PCA and LGM learning modules, where the first row denotes the false-color images of the first three PCs and the second row denotes the false-color images of LGMs. (a) SD. (b) IP. (c) KSC. (d) UP.

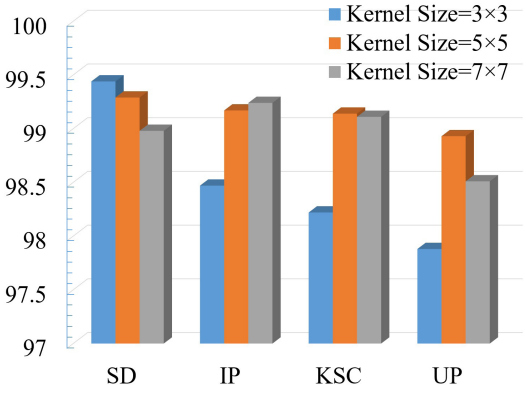

Fig. 11. Performance of CGCNNs with different sizes of guided kernel on different data sets.

number of GFEUs to 5. In the second experiment, the number of GFEUs is set to 3, 4, and 5 while fixing the size of guided kernel to 5×5. The results are illustrated in Figs. 11 and 12.

As can be seen from Fig. 11, the accuracy of CGCNN on SD decreases as the size of the guided kernel increases. On the contrary, the accuracy of CGCNN on IP increases along with

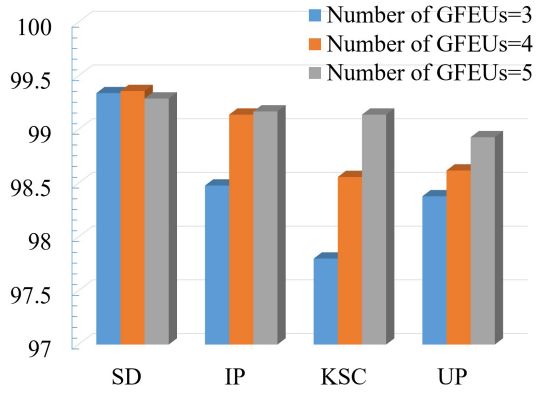

Fig. 12. Performance of CGCNNs with different numbers of GFEUs on different data sets.

TABLE X
EFFECTIVE ANALYSIS OF DENSE CONNECTION (OA%)

Dataset	Without Dense Connection	With Dense Connection
SD	96.57±1.39	**99.29±0.29**
IP	98.27±0.31	**99.17±0.15**
KSC	98.82±0.64	**99.14±0.14**
UP	98.83±0.35	**98.93±0.25**

the increasing size of the guided kernel. This phenomenon may be caused by the noises existing in them, i.e., the larger guided kernel could suppress more noise, while smaller guided kernel could preserve more small objects (details). In the KSC and UP data sets, the CGCNN with kernel size equal to 5×5 obtains the best performance. Taken together, we fix the guided kernel size to 5×5 in all experiments.

In the second experiment, as shown in Fig. 12, the OA indices of IP, KSC, and UP increase as the number of GFEUs increases. On the SD, the CGCNNs with different numbers of GFEUs show similar performance. Due to the limitations of the computational capacity of hardware (e.g., memory), we fix the number of GFEUs to 5 in all experiments.

F. Effectiveness of Dense Connection

The dense connection of deep networks is used to avoid overfitting. To demonstrate the effectiveness of dense connection for boosting the classification performance, we compare overall classification accuracies of CGCNN with and without dense connection. The experimental configuration is the same as Section III-C. The compared results are shown in Table X.

It is obvious that the dense connection is beneficial for improving the classification performance of CGCNN. The standard deviations of overall accuracies obtained by CGCNN with dense connection are all lower than that without dense connection, which indicates that the dense connection not only improves the classification accuracies but also enhances the classification stability.

G. Comparison of Program Performance

In this section, we compute the number of parameters of each deep-learning model, and the results are summarized

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TABLE XI
COMPARISON OF RUNNING TIME AND MODEL SIZES OF DIFFERENT METHODS ON DIFFERENT DATA SETS

Dataset	Time	2D-CNN [26]	TC-CNN [34]	3D-CNN [33]	MC-CNN [36]	SSRN [37]	HybridSN [35]	CGCNN
SD	Training	20.01s	35.49s	27.06s	34.64s	110.36s	91.32s	347.73s
	Testing	5.77s	5.75s	5.58s	6.71s	11.62s	9.76s	**0.43s**
IP	Training	40.20s	68.30s	50.74s	59.79s	245.33s	239.82s	221.26s
	Testing	3.67s	3.83s	3.54s	4.01s	7.51s	6.89s	**0.28s**
KSC	Training	19.24s	33.86s	19.99s	21.79s	69.19s	56.48s	2739.46s
	Testing	58.14s	63.75s	55.54s	60.86s	116.57s	98.24s	**3.42s**
UP	Training	22.71s	40.52s	30.37s	36.59s	120.78s	97.12s	1493.17s
	Testing	30.91s	28.81s	29.93s	35.79s	61.24s	52.60s	**1.97s**
Number of Parameters		68.38K	63.20K	101.15K	164.35K	152.26K	5104.17K	**59.75K**

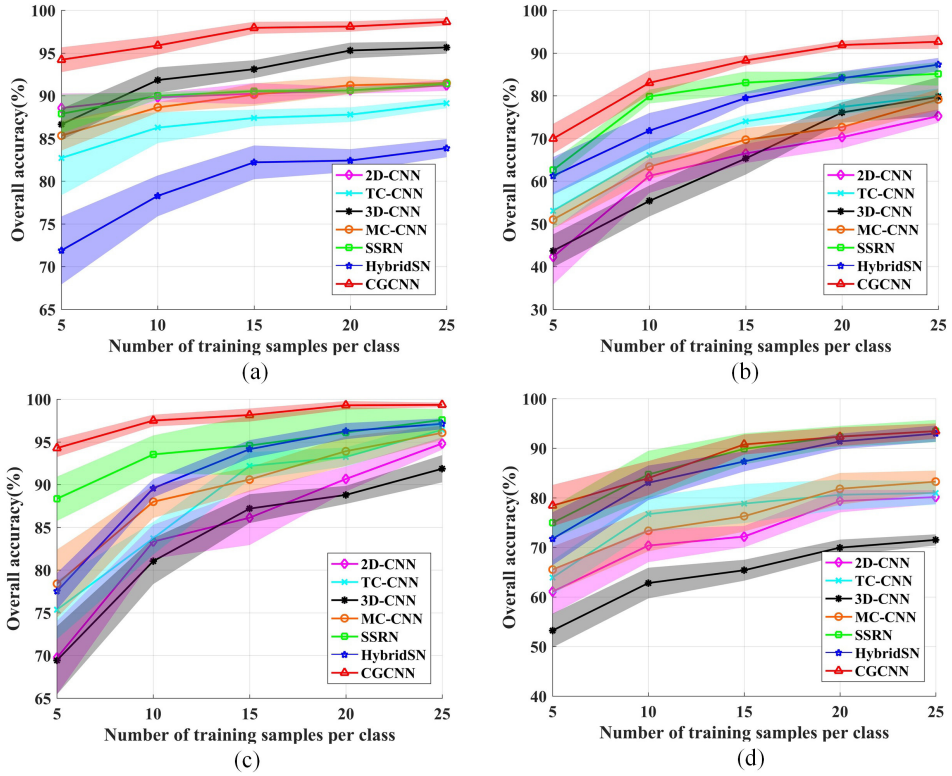

Fig. 13. Strip chart of classification results of different methods under different scales of training samples, where solid line denotes the mean and the shadow denotes the standard deviation. (a) SD. (b) IP. (c) KSC. (d) UP.

in Table XI. Our model has 50% less or fewer parameters than 3-D-CNN, MC-CNN, SSRN, and HybridSN. However, it gives the best results in general, which indicates that a reasonably structured network model is more effective than an overparameterized network for limited training samples.

In the industry field, because a trained model would be run thousands of times, the testing time of a method is far more important than the training time. For evaluating the program performance of the CGCNN, we record the training time and testing time of each method on each data set, as shown in Table XI. Note that the testing time here denotes the time of classifying all pixels in an HSI. Due to the network architecture that classifies pixels in parallel, the CGCNN also possesses high throughput capability, which indicates that the CGCNN is promising in large-scale HSI classification tasks and real-time HSI classification applications.

H. Performance of Small-Sample Learning

In real remote sensing applications, labeling the land-cover class of each pixel is expensive, and the training samples may not be sufficient. In order to investigate the performances of CGCNN in very limited training samples case, in this section, we test the OA indices of different methods with small training samples. We randomly select 5, 10, 15, 20, and 25 samples per class as the training data set, and one sample per class as the validation data set. The experiment is repeated ten times, and the mean and the standard deviation of OA indices are given in Fig. 13.

In the strip chart of small-sample learning results, it is clear that the proposed CGCNN shows satisfactory classification performance under different numbers of training samples and outperforms all the compared methods. Particularly, on the SD and KSC data sets, CGCNN obtains almost 95% overall classification accuracy with only five

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samples per class, which indicates that the proposed method has excellent small-sample learning ability.

IV. Conclusion

Because of the inherent drawback of shape-fixed convolution kernels, many CNN-based HSI classification methods tend to misclassify pixels in the cross-classes edges. By dynamically adjusting the kernel shape according to the HSI content, the proposed content-guided convolution could suppress the unexpected features of irrelevant classes and extract useful features in the feature learning of cross-classes regions. With the content-guided convolution, a novel GFEU is constructed to extract spectral–spatial features jointly with fewer parameters. By stacking multiple GFEUs with dense connections to establish the final classification framework, the CGCNN shows remarkable performance and could preserve the details of the land-cover classification map. In the future, inspired by the potential of the content-guided convolution, we will extend it to other sophisticated tasks, such as target detection and semantic segmentation of remote sensing images.

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References

[1] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[2] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[3] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[4] A. Signoroni, M. Savardi, A. Baronio, and S. Benini, "Deep learning meets hyperspectral image analysis: A multidisciplinary review," *J. Imag.*, vol. 5, no. 5, p. 52, May 2019.

[5] W. Liu, J. E. Fowler, and C. Zhao, "Spatial logistic regression for support-vector classification of hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 3, pp. 439–443, Mar. 2017.

[6] M. Pal and P. M. Mather, "An assessment of the effectiveness of decision tree methods for land cover classification," *Remote Sens. Environ.*, vol. 86, no. 4, pp. 554–565, Aug. 2003.

[7] C. Li, Y. Ma, X. Mei, C. Liu, and J. Ma, "Hyperspectral image classification with robust sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 641–645, May 2016.

[8] N. Huang and L. Xiao, "Hyperspectral image clustering via sparse dictionary-based anchored regression," *IET Image Process.*, vol. 13, no. 2, pp. 261–269, Feb. 2019.

[9] D. Lunga, S. Prasad, M. M. Crawford, and O. Ersoy, "Manifold-learning-based feature extraction for classification of hyperspectral data: A review of advances in manifold learning," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 55–66, Jan. 2014.

[10] H. Yuan and Y. Y. Tang, "Spectral–spatial shared linear regression for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 47, no. 4, pp. 934–945, Apr. 2017.

[11] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[12] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2008.

[13] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.

[14] J. Xia, J. Chanussot, P. Du, and X. He, "(Semi-) supervised probabilistic principal component analysis for hyperspectral remote sensing image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2224–2236, Jun. 2014.

[15] B.-C. Kuo, H.-H. Ho, C.-H. Li, C.-C. Hung, and J.-S. Taur, "A kernel-based feature selection method for SVM with RBF kernel for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 317–326, Jan. 2014.

[16] M. Pal, A. E. Maxwell, and T. A. Warner, "Kernel-based extreme learning machine for remote-sensing image classification," *Remote Sens. Lett.*, vol. 4, no. 9, pp. 853–862, Sep. 2013.

[17] Q. Wang, Y. Gu, and D. Tuia, "Discriminative multiple kernel learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 7, pp. 3912–3927, Jul. 2016.

[18] X. Cao, L. Xu, D. Meng, Q. Zhao, and Z. Xu, "Integration of 3-dimensional discrete wavelet transform and Markov random field for hyperspectral image classification," *Neurocomputing*, vol. 226, pp. 90–100, Feb. 2017.

[19] P. Ghamisi, J. A. Benediktsson, and M. O. Ulfarsson, "Spectral–spatial classification of hyperspectral images based on hidden Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2565–2574, May 2014.

[20] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.

[21] X. Huang, Q. Lu, L. Zhang, and A. Plaza, "New postprocessing methods for remote sensing image classification: A systematic study," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7140–7159, Nov. 2014.

[22] X. Huang *et al.*, "Multiple morphological profiles from multicomponent-base images for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 12, pp. 4653–4669, Dec. 2014.

[23] Y. Qian, M. Ye, and J. Zhou, "Hyperspectral image classification based on structured sparse logistic regression and three-dimensional wavelet texture features," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 4, pp. 2276–2291, Apr. 2013.

[24] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1–9.

[25] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.

[26] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[27] J. Yang, Y. Zhao, J. C.-W. Chan, and C. Yi, "Hyperspectral image classification using two-channel deep convolutional neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 5079–5082.

[28] H. Zhang, Y. Li, Y. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery using a dual-channel convolutional neural network," *Remote Sens. Lett.*, vol. 8, no. 5, pp. 438–447, May 2017.

[29] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[30] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[31] Y. Chen, Z. Lin, Z. Xing, W. Gang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[32] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jul. 2015.

[33] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[34] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

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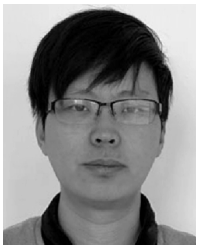
[35] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.
[36] C. Chen, J.-J. Zhang, C.-H. Zheng, Q. Yan, and L.-N. Xun, "Classification of hyperspectral data using a multi-channel convolutional neural network," in *Proc. Int. Conf. Intell. Comput.* Cham, Switzerland: Springer, Jul. 2018, pp. 81–92.
[37] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.
[38] W. Wang, S. Dou, Z. Jiang, and L. Sun, "A fast dense spectral–spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 1068, Jul. 2018.
[39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[40] G. Huang, Z. Liu, L. V. D. Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.
[41] J. Song, L. Xiao, M. Molaei, and Z. Lian, "Multi-layer boosting sparse convolutional model for generalized nuclear segmentation from histopathology images," *Knowl.-Based Syst.*, vol. 176, pp. 40–53, Jul. 2019.
[42] J. Song, L. Xiao, and Z. Lian, "Contour-seed pairs learning-based framework for simultaneously detecting and segmenting various overlapping cells/nuclei in microscopy images," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5759–5774, Dec. 2018.
[43] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1251–1258.
[44] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.
[45] M. Jaderberg *et al.*, "Spatial transformer networks," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2015, pp. 2017–2025.
[46] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, no. 6088, pp. 533–536, Oct. 1986.
[47] M. Abadi *et al.*, "TensorFlow: A system for large-scale machine learning," in *Proc. 12th USENIX Symp. Operating Syst. Design Implement. (OSDI)*, Nov. 2016, pp. 265–283.
[48] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. IEEE Int. Conf. Mach. Learn.*, Jul. 2015, pp. 448–456.
[49] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. IEEE Int. Conf. Learn. Represent.*, May 2015, pp. 1–15.



Qichao Liu (Student Member, IEEE) received the B.S. degree in computer science and technology from the Nanjing University of Science and Technology, Nanjing, China, in 2015, where he is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering.

His fields of interest include deep learning and remote sensing image processing.

Liang Xiao (Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from the Nanjing University of Science and Technology (NJUST), Nanjing, China, in 1999 and 2004, respectively.

From 2006 to 2008, he was a Post-Doctoral Research Fellow with the Pattern Recognition Laboratory, NJUST. From 2009 to 2010, he was a Post-Doctoral Fellow with the Rensselaer Polytechnic Institute, Troy, NY, USA. Since 2013, he has been the Deputy Director of the Jiangsu Key Laboratory of Spectral Imaging Intelligent Perception, Nanjing. Since 2014, he has been the Vice-Director of the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, NJUST, where he is currently a Professor with the School of Computer Science and Engineering. His fields of interests include remote sensing image processing, image modeling, computer vision, machine learning, and pattern recognition.

Jingxiang Yang (Member, IEEE) received the joint Ph.D. degree in control theory and control engineering and engineering science from Northwestern Polytechnical University, China, and Vrije Universiteit Brussel, Brussels, Belgium, in 2019.

He is currently a Lecturer with the Nanjing University of Science and Technology, Nanjing, China. His research interests include deep learning and its applications in hyperspectral image processing.

Jonathan Cheung-Wai Chan received the Ph.D. degree from The University of Hong Kong, Hong Kong, in 1999.

From 1998 to 2001, he was with the Department of Geography, University of Maryland, College Park, MD, USA. From 2001 to 2005, he was with the Interuniversity Micro-Electronics Center, Leuven, Belgium. From 2005 to 2011, he was with the Department of Geography, Vrije Universiteit Brussel (VUB), Brussels, Belgium. From 2013 to 2014, he was a Marie Curie Fellow with Fondazione Edmund Mach, San Michele all'Adige, Italy. He is currently a Guest Professor with the Department of Electronics and Informatics, VUB. He is also a Visiting Professor with the Warsaw University of Life Sciences, Warsaw, Poland, and Northwestern Polytechnical University, Xi'an, China.

Dr. Chan was a Guest Editor of *Remote Sensing* for the special issue on Spatial Enhancement of Hyperspectral Data and Applications.

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